Solution

Matrix Theory, Quiz 2C (2D) January 30, 2026

For the system of linear equations

$$\begin{aligned} 2x_1 \quad +2x_3+x_4 &= -5 \\ x_2 \ +x_3-x_4 &= 0 \\ x_1+x_2+2x_3 \quad &= -3 \end{aligned}$$

perform the following:

1 (a) Write down the augmented matrix.

5 (b) Use elementary row operations to reduce the matrix of part (a) into reduced row echelon form.

4 (c) Write the solution to the original system of equations in parametric vector form.

(a) $\left[\begin{array}{cccc|cc} 2 & 0 & 2 & 1 & -5 & 1 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 1 & 1 & 2 & 0 & -3 & 2 \end{array}\right]$

(b) $R_1 \leftrightarrow R_3 \left[\begin{array}{cccc|cc} 1 & 1 & 2 & 0 & -3 & 2 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 2 & 0 & 2 & 1 & -5 & 1 \end{array}\right] \begin{array}{l} -2 \\ \\ \\ \end{array} \xrightarrow{R_3-2R_1} \left[\begin{array}{cccc|cc} 1 & 1 & 2 & 0 & -3 & 2 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & -2 & -2 & 1 & 1 & -3 \end{array}\right] \begin{array}{l} \\ \times 2 \\ \\ \end{array}$

$\xrightarrow[R_3+2R_2]{R_1-R_2} \left[\begin{array}{cccc|cc} 1 & 0 & 1 & 1 & -3 & 0 \\ 0 & 1 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{array}\right] \begin{array}{l} \\ \\ -1 \end{array} \xrightarrow[\substack{R_2-R_3 \\ -R_3}]{R_1+R_3} \left[\begin{array}{cccc|cc} 1 & 0 & 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 & -1 \end{array}\right]$

(c) Version 2C : $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -2-x_3 \\ -1-x_3 \\ x_3 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \\ -1 \end{pmatrix} + x_3 \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}$

Version 2D : $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1-x_3 \\ 1-x_3 \\ x_3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ -1 \end{pmatrix} + x_3 \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}$